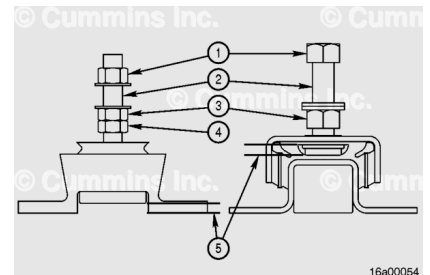


Trainings ▾

General Information

This illustration shows the different types of marine isolators, and the location of the snubber and corresponding gap used to determine loading of the isolator.

1. Top nut
2. Leveling stud
3. Adjusting nut
4. Jam nut
5. Snubber gap.



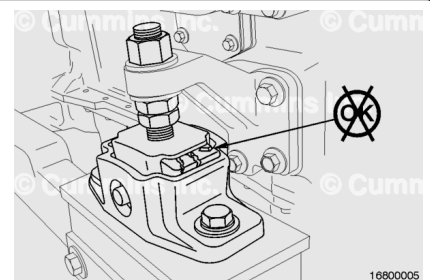
Generally, it is recommended to replace isolators as a set, but the age of the isolators and the conditions that caused the damage need to be reviewed.

Isolators of different designs should **not** be mixed in the same installation.

Initial Check

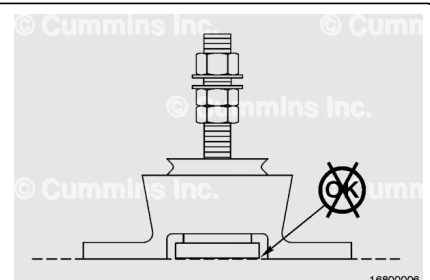
Check that the rubber portion of the isolator is **not** cracked, damaged, or pushed out from the isolator housing.

If the rubber is damaged, the isolator will need to be replaced.



Check that the isolator is **not** fully compressed. If an isolator is in the fully compressed condition, there will be no clearance in the snubber gap.

If **only** one isolator is compressed, then the loading of the isolators may be unbalanced. Reference the Install Section of this procedure for steps to balance the load. If the isolator is



still compressed after adjusting the load, then the isolator will need to be replaced.

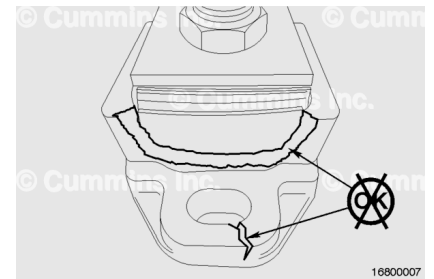
If two isolators that are diagonally across from each other are significantly more compressed or fully compressed, then the loading on the isolators is unbalanced. Reference the Install Section of this procedure for steps to balance the load.

If two isolators that are both on one side of the engine or both at the front or rear of the engine are fully compressed, then the isolators will need to be replaced.

Check that the metal parts of the isolator are **not** damaged, cracked, bent, or warped. If they are, the isolator will need to be replaced.

Check for excessive corrosion. If corrosion is enough to weaken the structure or impede the function of the isolator, it will need to be replaced.

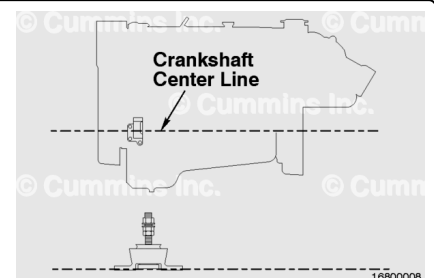
If an isolator is damaged or worn and less than one year old, check that the isolator part number is correct for the engine.



Check that the isolator mounting base is parallel with the engine crankshaft centerline and that the stud is perpendicular to the base, when looking from the side of the engine. The oil pan flange can be used as a visual reference. The engine support brackets **must** also be parallel.

The alignment of the isolator base to the crank centerline should be within four degrees of parallel.

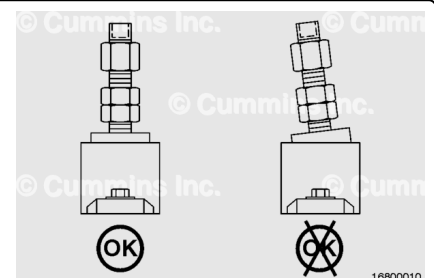
The alignment of the isolator stud to the base should be within four degrees of perpendicular.



Check that the isolator mounting base is parallel to the transverse crankshaft centerline and that the stud is perpendicular to the base when looking from the front or back of the engine. The engine support brackets **must** also be parallel.

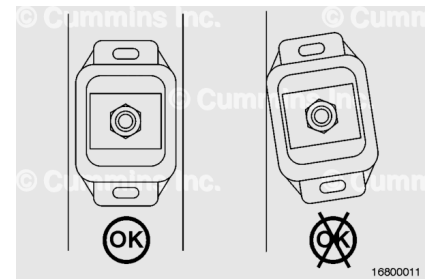
The alignment of the isolator base to the transverse crankshaft centerline should be within two degrees of parallel.

The alignment of the isolator stud to the base should be within two degrees of perpendicular.

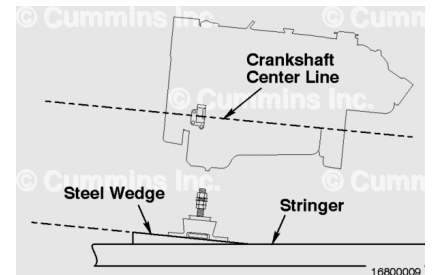


Check that the isolator mounting base is parallel to the crankshaft centerline when looking from the top of the engine.

The alignment of the isolator base to the crankshaft centerline should be within two degrees of parallel.



If the isolator is out of alignment, the mounting will need to be adjusted. Wedges can be used to shim the base to achieve proper alignment. Shims **must** be made of a solid material that will **not** compress under the weight of the engine.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Disconnect the battery. Refer to the original equipment manufacturer (OEM) service manual.
- Disconnect the propeller shaft. Disengage the flange. Refer to Procedure 016-025 in Section 16. (</qs3/pubsys2/xml/en/procedures/00/00-016-025.html>)

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this

component or assembly.

The engine needs to be supported before removing the isolators. The amount of space available will determine the best method for this support.

Remove the top nut from the isolator's adjusting stud. Remove the bolts or capscrews retaining the isolator base.

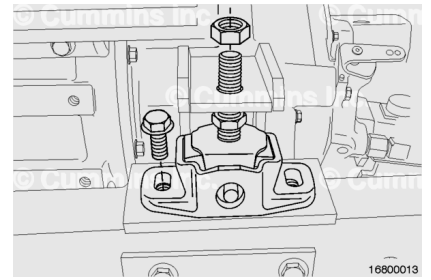
Raise the engine if necessary to facilitate the removal of the isolator.

Make sure that any equipment used for hoisting or jacking the engine is capable of handling the weight of the engine and marine gear. Any engine component used for attaching or support such as the lifting brackets or engine supports **must** be correct for the engine. Reference the appropriate manual for General Engine information in Section V for the engine weight.

Remove the isolator.

If reusing the isolator, mark the position of the isolator relative to the engine to make sure of installation in the same location.

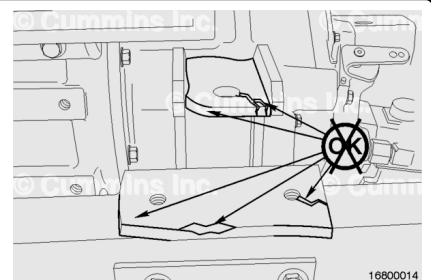
If replacing the worn or damaged isolator with a new isolator, measure the height of the adjusting nut on the stud to assist in setting the new isolator adjusting nut height.



Inspect for Reuse

Inspect the vessel stringers or engine bed to make sure they have **not** been damaged and that they can continue to support the weight of the engine. This is especially true of stringers constructed of wood, wood core, or hollow fiberglass.

Inspect the mounting holes in the stringers or the engine bed for damage. If the stringers, engine bed, or mounting holes need to be repaired, contact an OEM qualified repair location.



Install

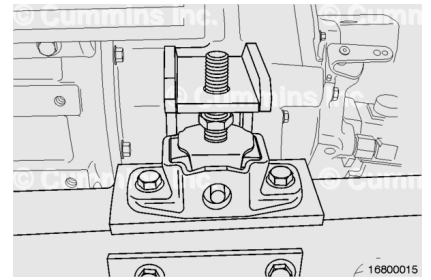
Set the new isolator in place. Check that the isolator is properly aligned. Reference the Initial Check section in this procedure for alignment information.

Set the adjusting nut at the approximate height of the isolator removed.

Replace other isolators that need to be changed using the same technique.

Install the washer and top nut.

Lower the engine so that its weight is fully supported by the isolators.



Verify that the isolators are loaded evenly. Depending on the configuration of the engine, the weight may **not** be evenly distributed on all four corners.

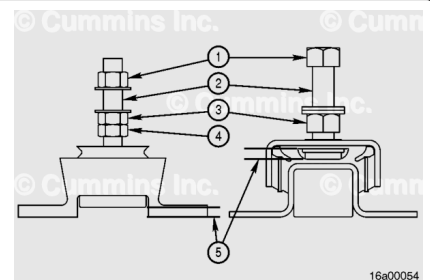
The engine will be heavier at either the front or back, depending on the configuration. Therefore, the isolators on the heavier end will be compressed more.

Check the snubber gap on all isolators.

If **only** one isolator or two isolators that are diagonally across from each other are significantly more compressed, the loading on the isolators is unbalanced.

Adjust the loading on the isolators by raising or lowering the adjusting nut on the stud. Raising the adjusting nut will increase the load. Lowering the adjusting nut will decrease the load.

The engine should be lifted and the load removed from the isolator before moving the adjusting nut, to prevent damage to the threads.



Finishing Steps

⚠ WARNING ⚠



Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Align the propeller shaft. Refer to Procedure 016-025 in Section 16. (</qs3/pubsys2/xml/en/procedures/00/00-016-025.html>)
- Connect the battery cables. Refer to the OEM service manual.
- Start the engine and check for proper operation.